

LambdaEta yield from production-vertex missing mass

Macro: plot_lambda_eta_yield.cc | Executed: 2026-10-01 08:35:15

Beam momentum setting: 715 MeV/c

Runs used: 02682, 02683, 02684, 02686, 02687, 02689, 02690, 02691, 02692, 02693

Input: /gpfs/home/had/haein/data/JPARC2025Nov_root/physics-715/runXXXXX_DstTPCLambdaEta.root

Selection: X_prod_found; Lambda mass window $1.09 < M(p \pi^-) < 1.13 \text{ GeV}/c^2$

Lambda mass displayed: 1.04 to 1.20 GeV/c^2 ; peak fit: 1.10 to 1.13 GeV/c^2

Pre-cut Lambda peak fit model: Gaussian + linear background, 1.100 to 1.130 GeV/c^2

Production vertex LH2 cut: off

Pion selection: lambda_pion_track_id matched to pion_beam_window_status == 0 when branch exists

Eta background fit: linear, 0.00 to 0.50 GeV/c^2

Eta yield: observed minus fitted background, 0.50 to 0.60 GeV/c^2

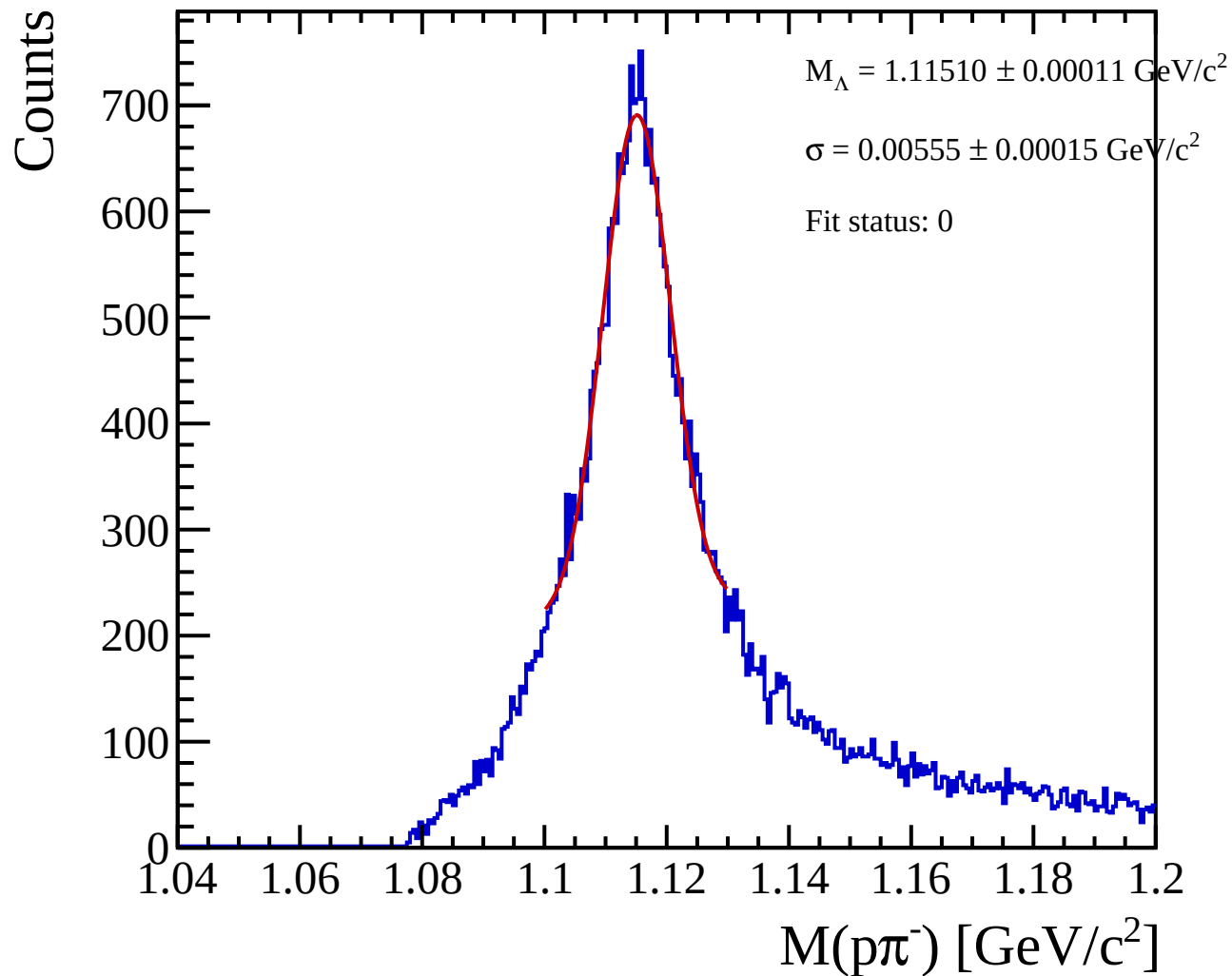
Scale scan: separate p and pi momentum factors 1.00, 1.05, 1.10, 1.15, 1.20

Scale histograms: accepted pion-veto candidates; no Lambda mass window; missing mass requires production

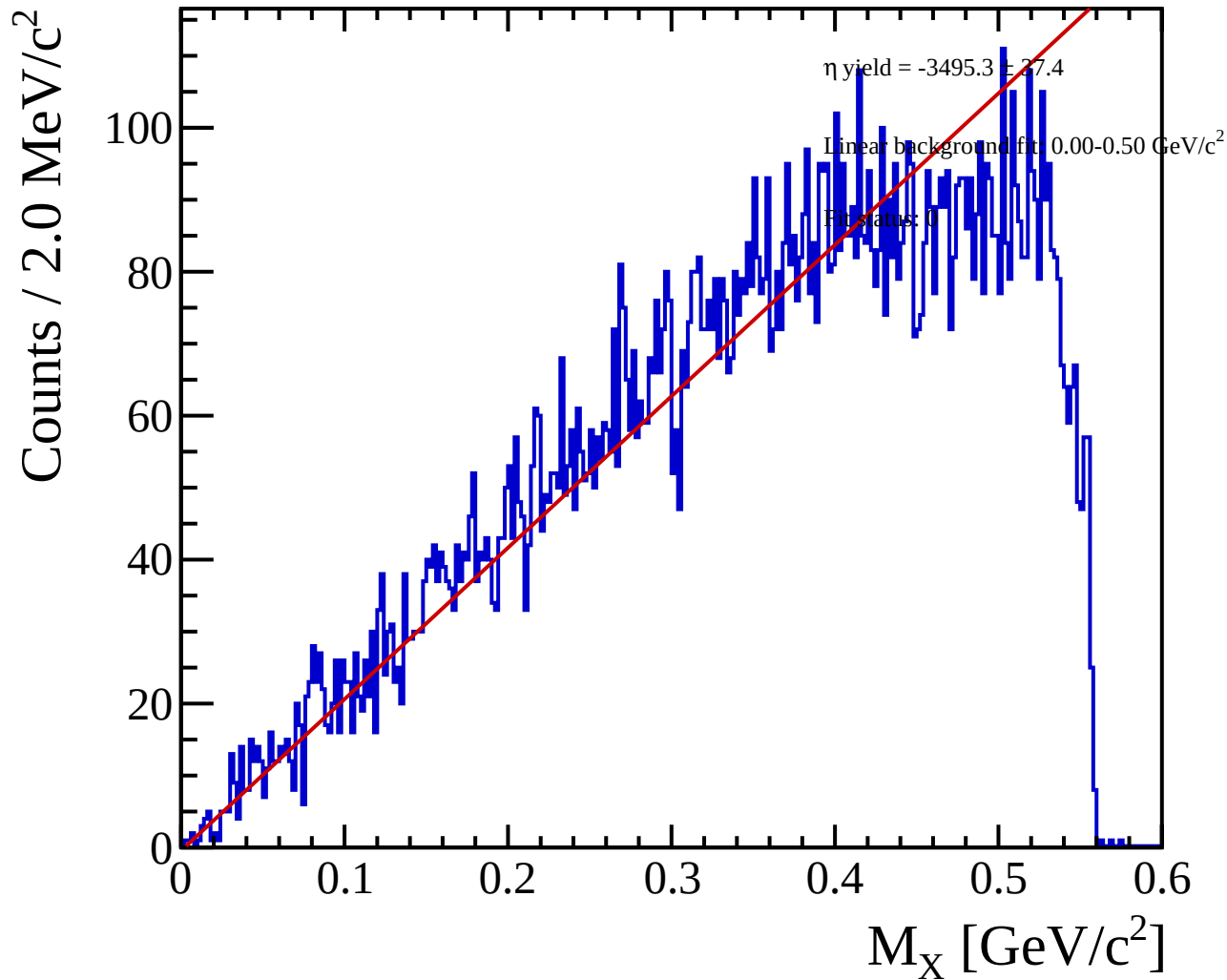
Scale peak fits: Lambda 1.10-1.13; eta signal 0.54-0.58 GeV/c^2

Events read: 1716788 | candidates checked: 54083 | pre-cut Lambda entries: 52409 | eta selection: 28951

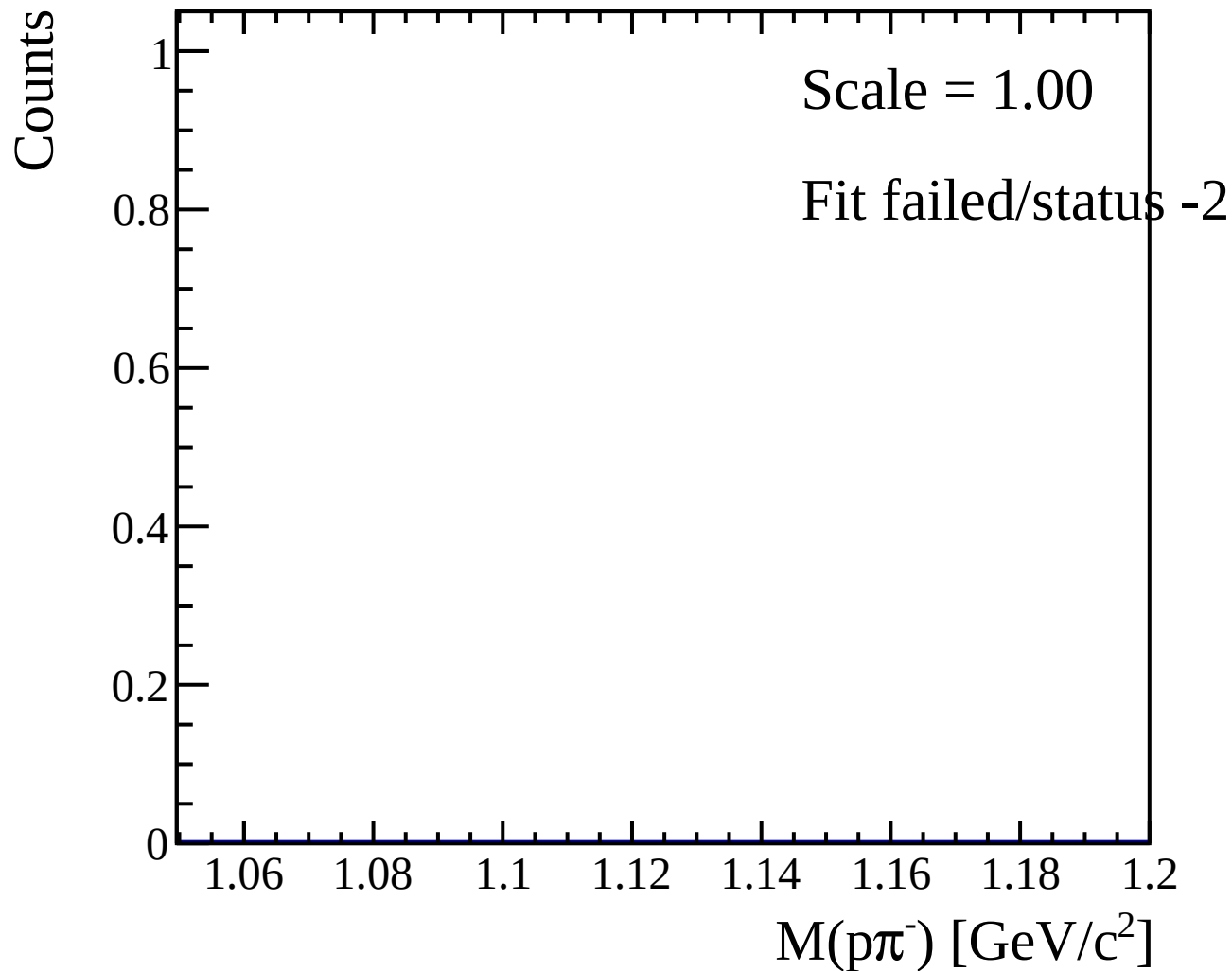
Λ invariant mass before Lambda-mass cut



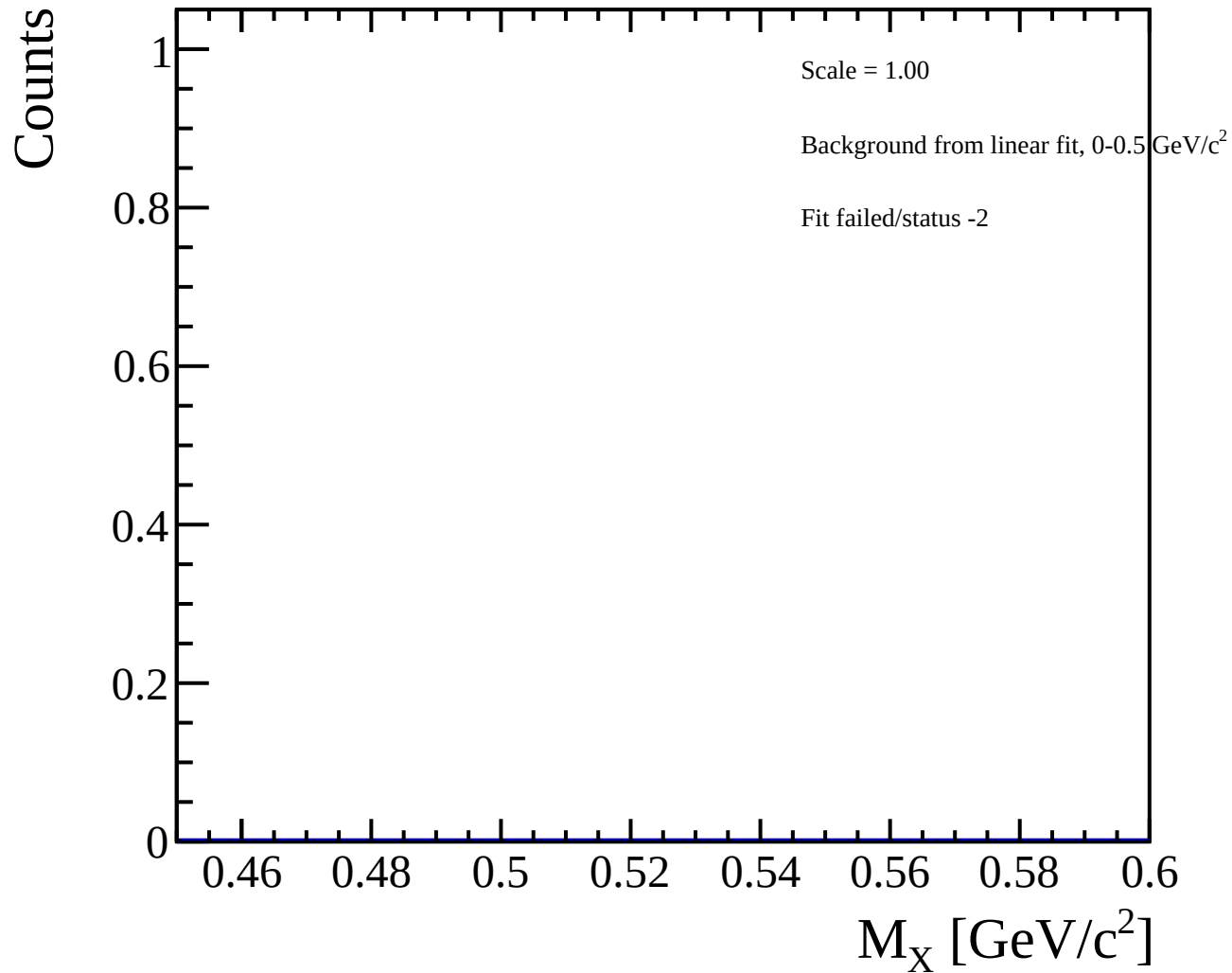
LambdaEta missing mass



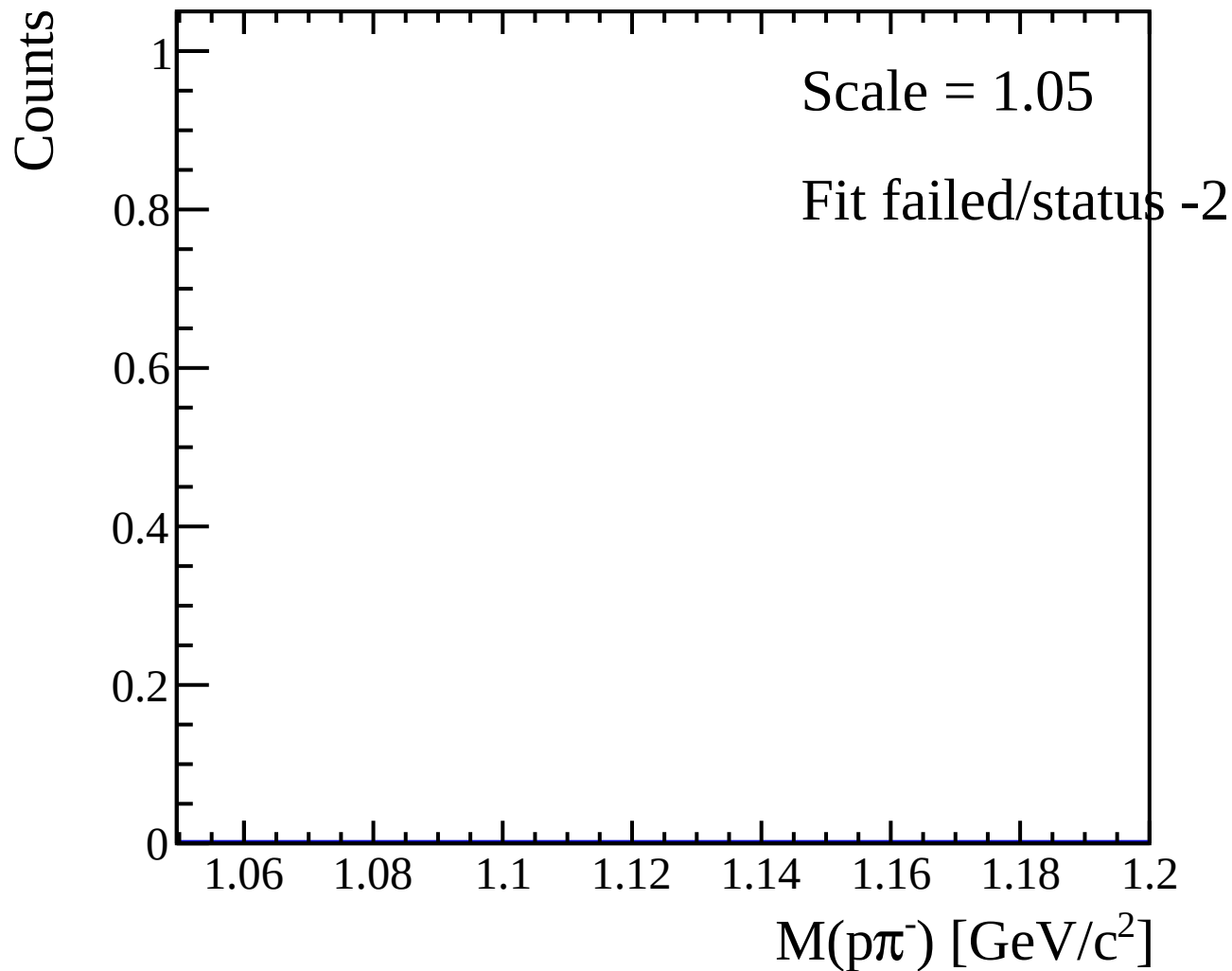
Scaled Λ mass, common daughter momentum scale = 1.00



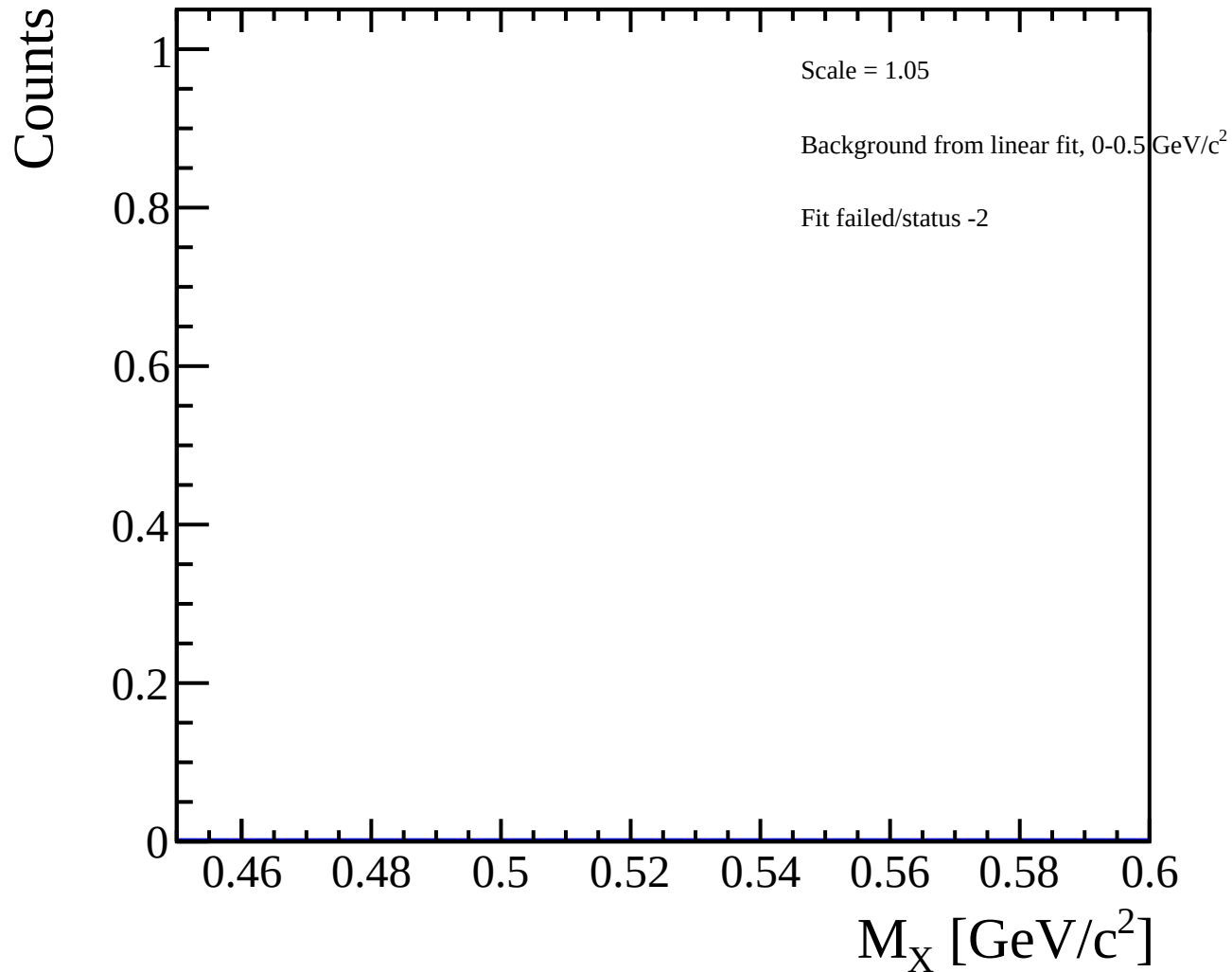
Scaled missing mass, common daughter momentum scale = 1.00



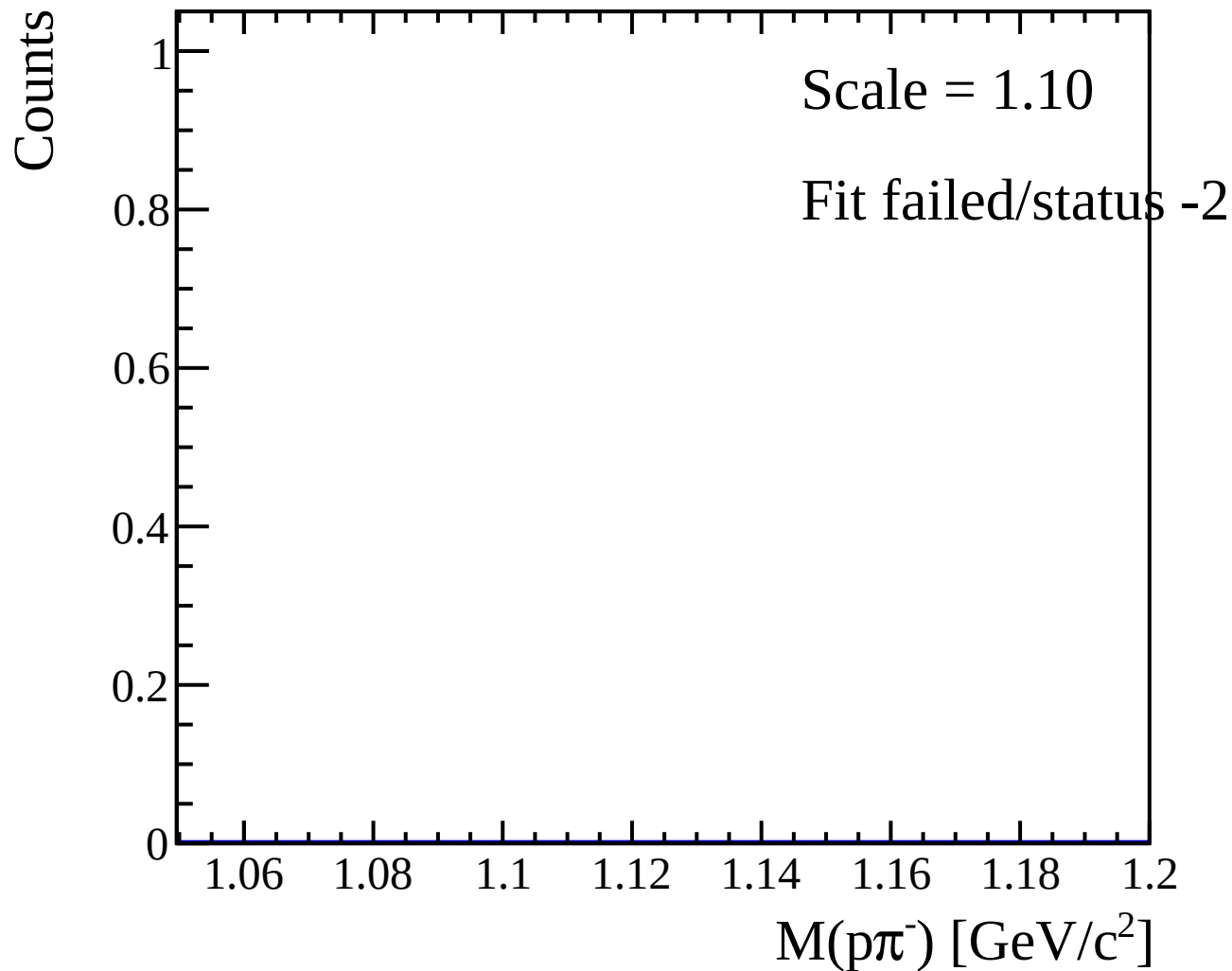
Scaled Λ mass, common daughter momentum scale = 1.05



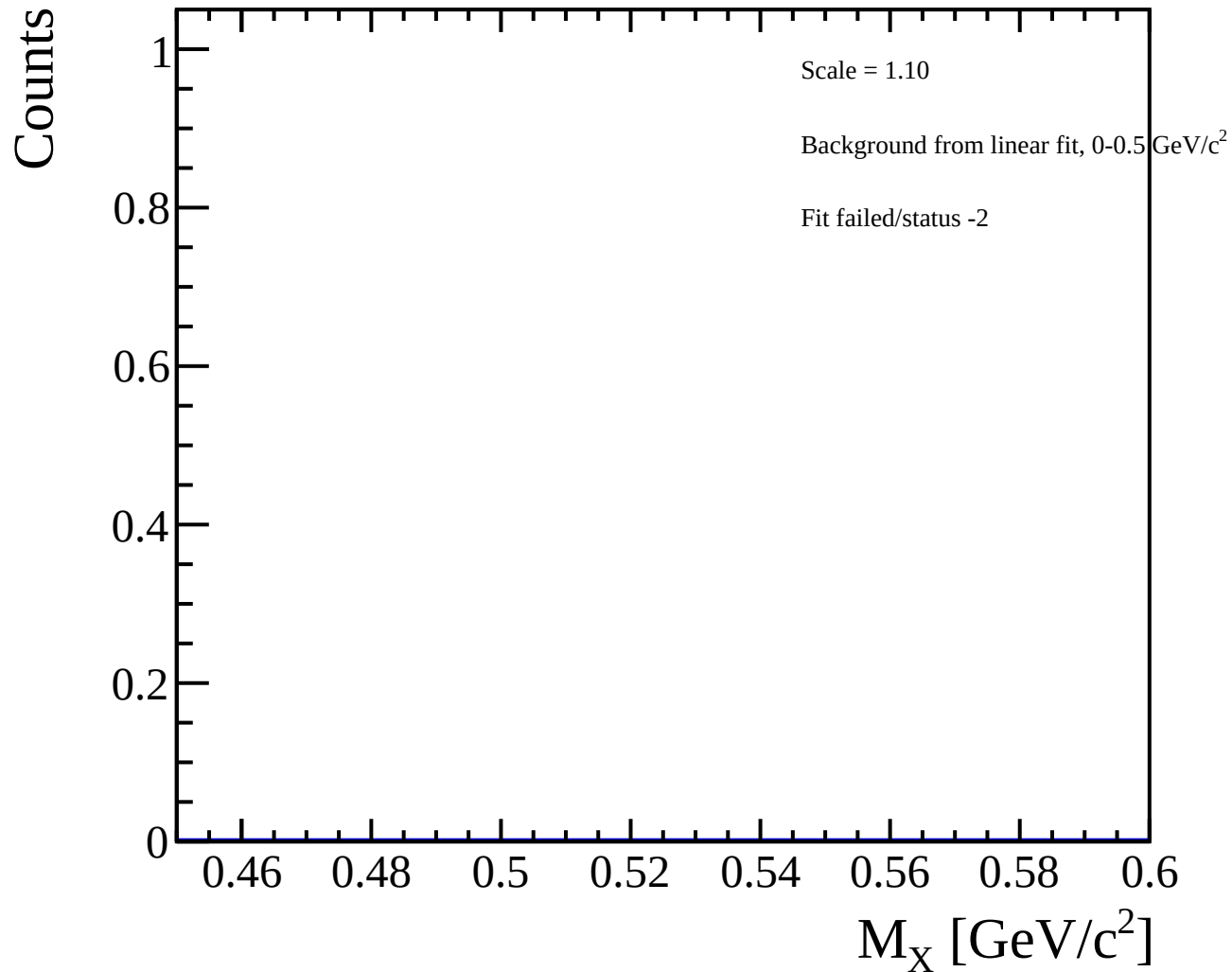
Scaled missing mass, common daughter momentum scale = 1.05



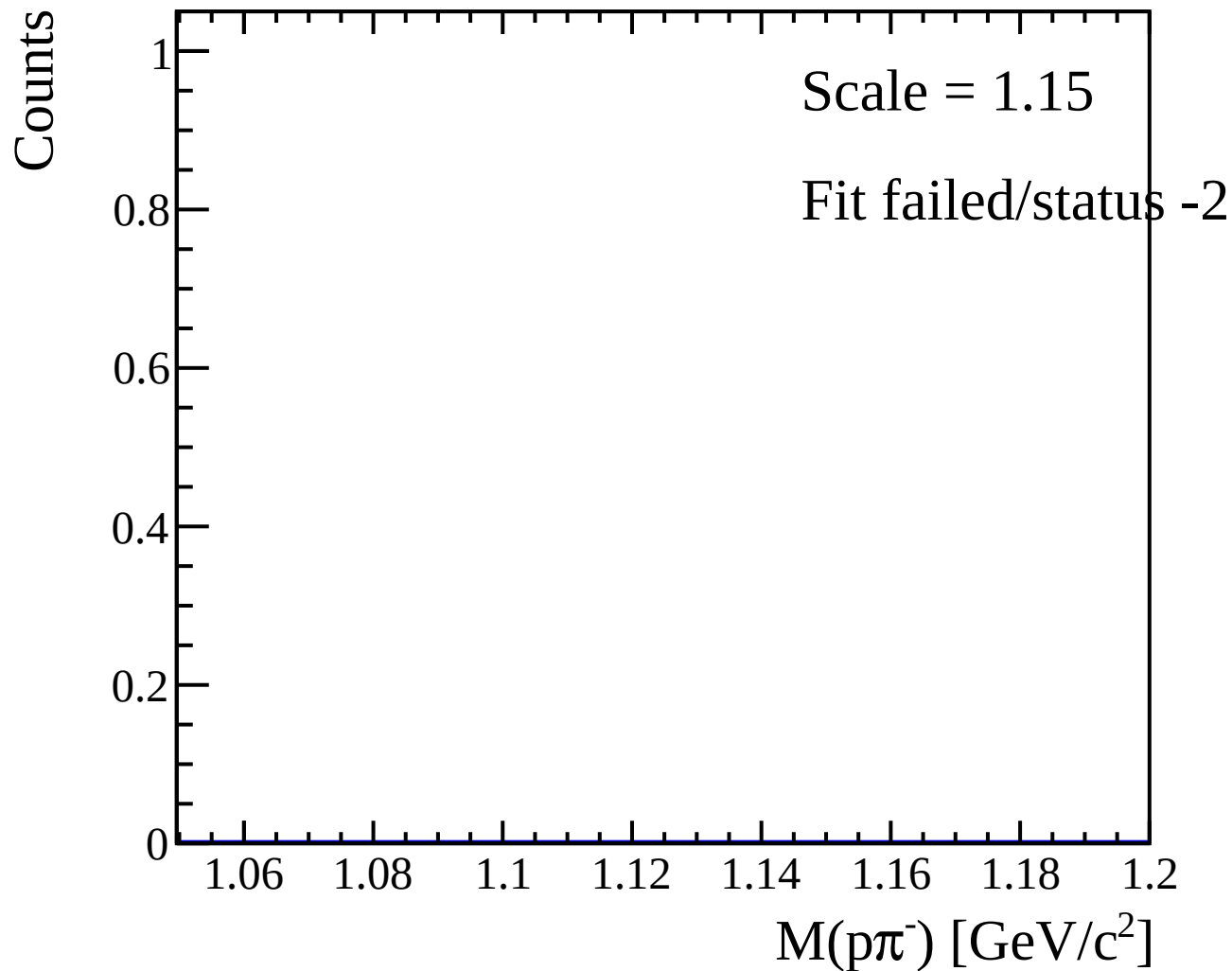
Scaled Λ mass, common daughter momentum scale = 1.10



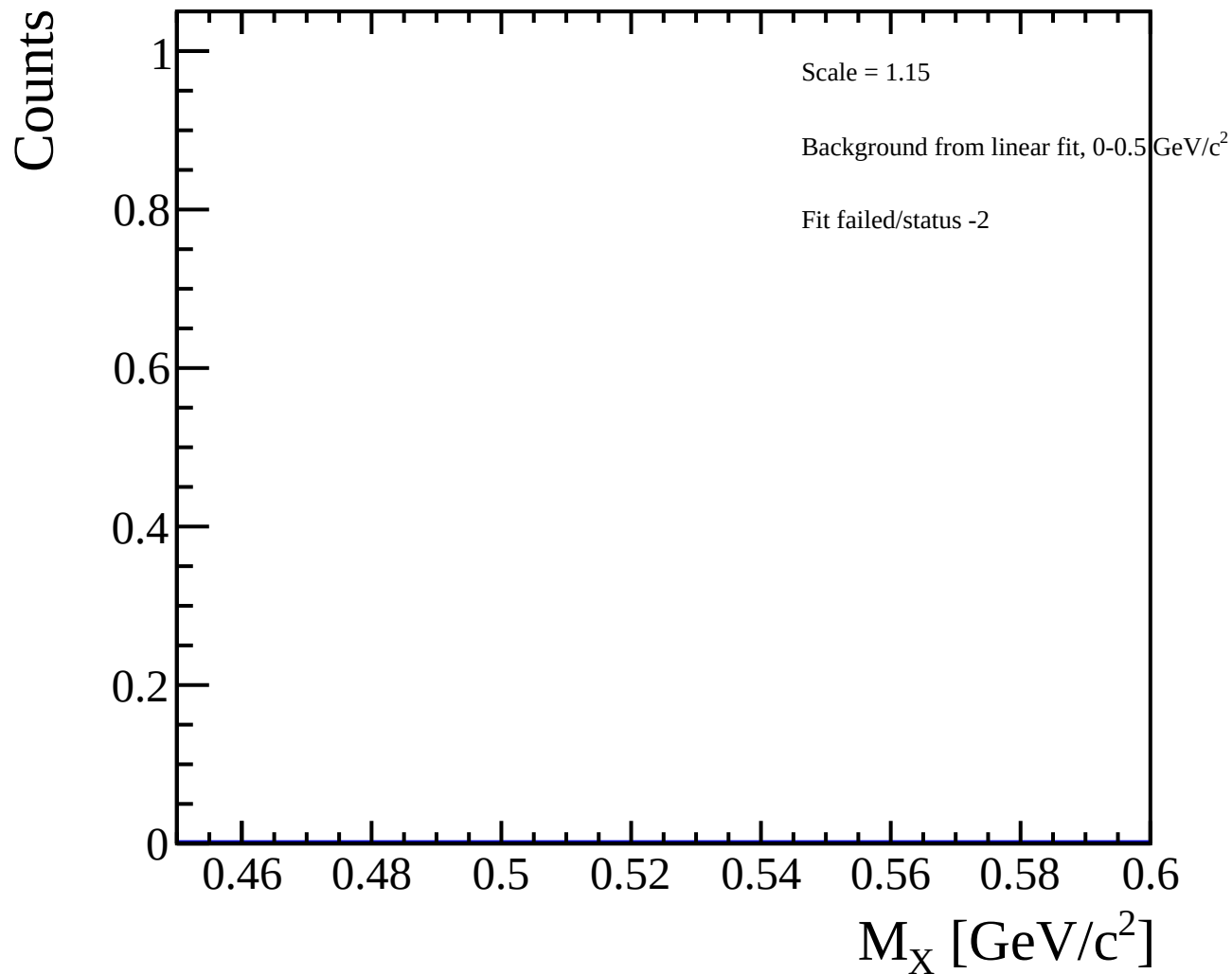
Scaled missing mass, common daughter momentum scale = 1.10



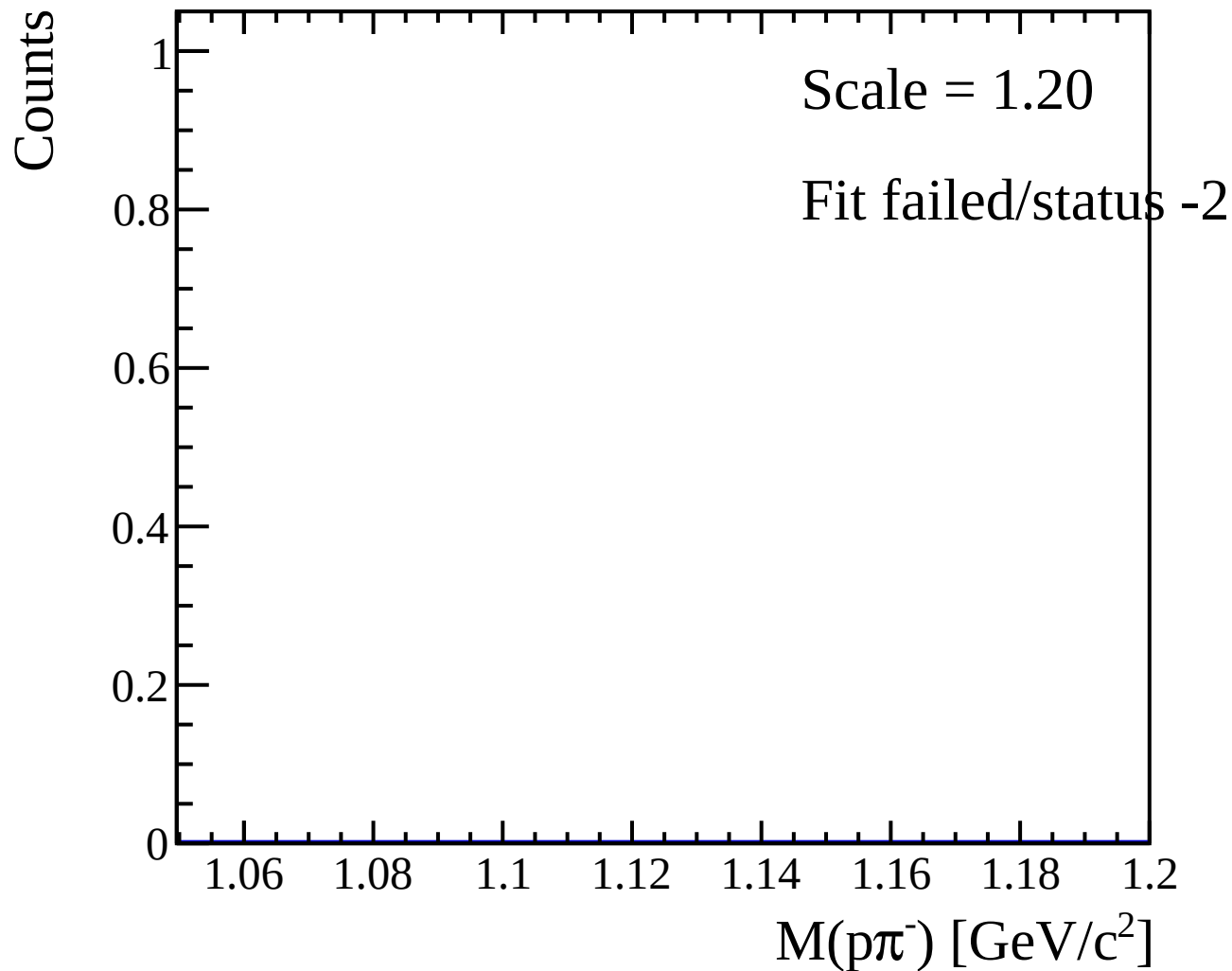
Scaled Λ mass, common daughter momentum scale = 1.15



Scaled missing mass, common daughter momentum scale = 1.15



Scaled Λ mass, common daughter momentum scale = 1.20



Scaled missing mass, common daughter momentum scale = 1.20

